

Letter

A single-cell atlas of CD19 chimeric antigen receptor T cells

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Autologous chimeric antigen receptor T cells directed toward CD19 (CART19) have significantly improved the outcomes of relapsed or refractory large B cell lymphoma (rrLBCL).¹ However, the majority of patients do not have durable responses; thus, strategies to improve outcomes are critically needed. The mechanisms underlying CART19 treatment failure are complex and likely interrelated. They may include the initial functional states and composition of T cells harvested during apheresis, the subsequent functional characteristics of the CART19 infusion product, and tumor-intrinsic properties such as the tumor's resistance to T cell infiltration and/or cytotoxicity. Careful examination of these T cell phenotypes using omics-based strategies such as single-cell RNA sequencing (scRNA-seq) has revealed characteristics associated with CAR T cell resistance (reviewed in Yang et al.²). scRNA-seq analysis of CAR T cell infusion products has highlighted important associations with memory, exhausted, and regulatory T cell states.^{3,4} However, the sample sizes in these studies have been limited, both in terms of the number of patients and the number of cells. Here we present a resource of scRNA-seq data from the infusion products of 59 rrLBCL patients (Table S1A) treated with standard-of-care axicabtagene ciloleucel (Axi-cel; 417,167 high-quality single cells) that we have made publicly available to facilitate ongoing and future discovery efforts that will improve patient outcomes. As a proof-of-principle for the utility of this dataset, we identified features that were significantly different between products from responders (complete response [CR]) and non-responders (stable disease,

partial response, and progressive disease) at 3-month followup by PET/CT, an important landmark for long-term outcomes.

A detailed description of the methods is provided in the [supplemental information](#). Cells from all 59 patients were included for unsupervised clustering to define major cell types. However, only 35 patients were included for outcome comparisons because 5 patients were not evaluable and 19 patients received additional therapy on a clinical trial following their infusion that likely modified their outcomes.⁵ Clustering of all cells identified previously reported ICANS-associated cells (IACs) and non-ICANS-associated myeloid-like cells,³ as well as CD4⁺, CD8⁺, and double-negative clusters (Figure S1A; Table S1B). To classify T cells, we employed a K-nearest neighbors approach⁴ to calculate smoothed expressions of CD3 (average of *CD3D*, *CD3E*, *CD3G*, and *CD247*), CD8 (average of *CD8A* and *CD8B*), and *CD4*. We stringently defined CD4⁺ and CD8⁺ cells using smoothed gene expressions (Figure S1B) and re-clustered these subsets to define transcriptionally distinct subclusters. The CD4 T cell clusters (Figure S1C; Table S1C) included T-regulatory (FOXP3-positive) and cytotoxic (cytotoxicity signature high) subsets that have previously been implicated in CAR T cell failure⁴ and durable response,⁶ respectively. However, the relative proportions of these and other CD4 T cell clusters were not significantly different between responders (R, n = 15) and non-responders (NR, n = 20) using scCODA⁷ (Figure S1D). Similarly, subclustering of CD8 T cells identified memory, effector, and dysfunctional clusters that have previously been

implicated in CAR T cell response³ (Figure S1E; Table S1D), but the frequencies of these clusters were not significantly different between responders and non-responders (Figure S1F). Beyond the previously described association with the IACs cluster,³ this approach also did not identify significant associations between CD4 or CD8 cell subclusters and the development of grade 3–4 ICANS (data not shown). We therefore explored broader transcriptional patterns of CD4 and CD8 T cells between groups. This revealed significant and reproducible patterns of differential gene expression across these cell lineages, within all cells (Figure S1G; Table S1E) and restricting to CAR-positive (Table S1F) CD4⁺ and CD8⁺ T cells, summarized by two major trends. First, cells from NR patients had significantly higher expression of genes and gene sets associated with cytokine signaling (over-representation analysis using hypergeometric test, FDR < 1e-5), including “cytokine signaling in the immune system,” “signaling by interleukins,” “chemokine receptors bind chemokines,” and “TNFs bind their physiological receptors” from the Reactome pathway database (Table S1G). Similar pathways were also associated with development of grade 3–4 ICANS (Table S1H and S1I). Second, cells from patients with CR showed significantly higher expression of glycolysis pathway genes, including genes encoding proteins that catalyze 10/11 steps in the metabolism of glucose into lactate (Figure S1G). These genes were by-in-large not associated with the development of grade 3–4 ICANS (Figure S1H; Table S1H). Furthermore, we evaluated the KEGG glycolysis and



glyconeogenesis pathway by preranked GSEA, which revealed a significant enrichment in CD4 and CD8 T cell compartments associated with achieving a CR (Figure S1I) but not with the development of high-grade ICANS (FDR Q-value > 0.9). Glycolysis pathway genes were similarly significant in both CAR-positive and CAR-negative populations and in both CD4 and CD8 T cell subsets (Figure S1J). To validate this, we investigated the expression of glycolysis pathway genes within the recently published Haradhvala et al. dataset⁴ (18 patients treated with Axi-cel, 9 R, and 7 NR; 38,875 cells) and confirmed a significantly higher level of these genes in responders (Figure S1J), despite having a relatively smaller statistical power due to a smaller number of patients. Using a more stringent approach employing pseudobulk comparisons, we aimed to define specific cell types for which glycolysis may be most important by comparing each gene across each CD4 and CD8 T cell clusters. While individual genes such as PGK1 tended to be significant across clusters, the clusters with the most reproducible change of genes within the full pathway were CD8 C2 and CD8 C4, aligning to memory (CD27, *LEF1*, and *TCF7*; Table S1D) and tissue resident memory (*CXCR6*, *SELPLG*, and *CCR8*; Table S1D) phenotypes, respectively (Figure S1K). Metabolic health has been directly implicated in T cell function by multiple studies. For example, metabolic switching from oxidative phosphorylation to glycolysis is associated with increased effector function⁸ and is reduced following PD-1 and CTLA-4 engagement⁹ and during the process of T cell exhaustion.¹⁰ Our data, therefore, suggest that the expression of glycolysis pathway enzymes and, by extension, the metabolic profile of infusion product T cells may be a factor underlying the heterogeneity in responses to CD19 CAR T cell therapy in rrLBCL but may be independent of important toxicities such as ICANS.

In conclusion, here we provide a large scRNA-seq repository of Axi-cel CD19 CAR T cell infusion products from rrLBCL and show proof of principle for the utility of this dataset by identifying a significant and reproducible overexpression of glycolysis pathway genes

in the CD4 and CD8 T cells from patients achieving a CR at their 3-month followup. The role of transcriptional activation of glycolysis pathway genes in CAR T cell function and the response of rrLBCL patients to CD19 CAR T cells require direct experimental validation. However, this observation serves to demonstrate the discovery potential of large single-cell datasets in CAR T cell therapy. Therefore, to facilitate access and drive future discoveries in the field, we have uploaded these data to the CellXGene repository (<https://cellxgene.cziscience.com/collections/d9a9760c-e43d-404d-86aa-447c9a1e9120>) and made matrices available through GEO (GSE241783) and raw data available through EGA (EGAS00001004576).

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.ccell.2023.08.015>.

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REFERENCES

- Neelapu, S.S., Locke, F.L., Bartlett, N.L., Lekakis, L.J., Reagan, P.M., Miklos, D.B., Jacobson, C.A., Braunschweig, I., Oluwole, O.O., Siddiqui, T., et al. (2021). Comparison of 2-year outcomes with CAR T cells (ZUMA-1) vs salvage chemotherapy in refractory large B-cell lymphoma. *Blood Adv.* 5, 4149–4155.
- Yang, J., Chen, Y., Jing, Y., Green, M.R., and Han, L. (2023). Advancing CAR T cell therapy through the use of multidimensional omics data. *Nat. Rev. Clin. Oncol.* 20, 211–228.
- Deng, Q., Han, G., Puebla-Orsorio, N., Ma, M.C.J., Strati, P., Chasen, B., Dai, E., Dang, M., Jain, N., Yang, H., et al. (2020). Characteristics of anti-CD19 CAR T-cell infusion products associated with efficacy and toxicity in patients with large B-cell lymphomas. *Nat. Med.* 26, 1878–1887.
- Haradhvala, N.J., Leick, M.B., Maurer, K., Gohil, S.H., Larson, R.C., Yao, N., Gallagher, K.M.E., Katsis, K., Frigault, M.J., Southard, J., et al. (2022). Distinct cellular dynamics associated with response to CAR-T therapy for refractory B cell lymphoma. *Nat. Med.* 28, 1848–1859.
- Strati, P., Jallouk, A., Deng, Q., Li, X., Feng, L., Sun, R., Adkins, S., Johncy, S., Cain, T., Steiner, R.E., et al. (2023). A Phase I Study of Prophylactic Anakinra to Mitigate ICANS in

- Patients with Large B-cell Lymphoma. *Blood Adv.* <https://doi.org/10.1182/bloodadvances.2023010653>.
6. Melenhorst, J.J., Chen, G.M., Wang, M., Porter, D.L., Chen, C., Collins, M.A., Gao, P., Bandyopadhyay, S., Sun, H., Zhao, Z., et al. (2022). Decade-long leukaemia remissions with persistence of CD4(+) CAR T cells. *Nature* 602, 503–509.
 7. Büttner, M., Ostner, J., Müller, C.L., Theis, F.J., and Schubert, B. (2021). scCODA is a Bayesian model for compositional single-cell data analysis. *Nat. Commun.* 12, 6876.
 8. Chang, C.H., Curtis, J.D., Maggi, L.B., Jr., Faubert, B., Villarino, A.V., O'Sullivan, D., Huang, S.C.C., van der Windt, G.J.W., Blagih, J., Qiu, J., et al. (2013). Posttranscriptional control of T cell effector function by aerobic glycolysis. *Cell* 153, 1239–1251.
 9. Patsoukis, N., Bardhan, K., Chatterjee, P., Sari, D., Liu, B., Bell, L.N., Karoly, E.D., Freeman, G.J., Petkova, V., Seth, P., et al. (2015). PD-1 alters T-cell metabolic reprogramming by inhibiting glycolysis and promoting lipolysis and fatty acid oxidation. *Nat. Commun.* 6, 6692.
 10. Bengsch, B., Johnson, A.L., Kurachi, M., Odorizzi, P.M., Pauken, K.E., Attanasio, J., Stelekati, E., McLane, L.M., Paley, M.A., Delgoffe, G.M., and Wherry, E.J. (2016). Bioenergetic Insufficiencies Due to Metabolic Alterations Regulated by the Inhibitory Receptor PD-1 Are an Early Driver of CD8(+) T Cell Exhaustion. *Immunity* 45, 358–373.